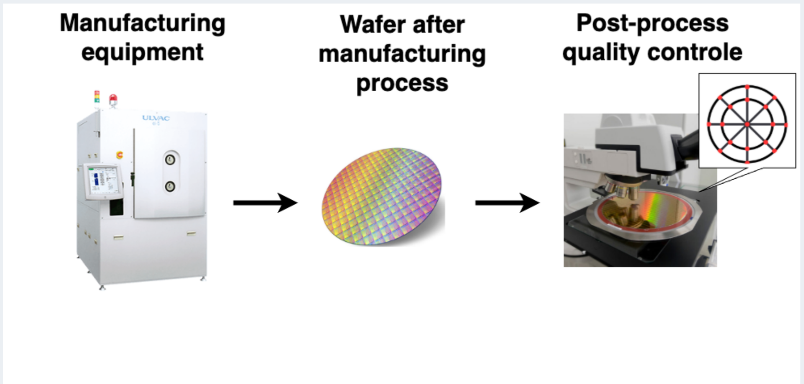
A Physics-Consistency Monitoring Framework for Virtual Metrology Systems

Amina Mević[1,2] Senka Krivić[1]
[1]University of Sarajevo – Faculty of Electrical Engineering ▪ [2]Infineon Technologies AG ▪ PHYCONBA 2026, Sarajevo

1. Virtual metrology for thickness monitoring



F1. After manufacturing, wafer quality is inspected at 17 characteristic points. VM uses ML models to predict quality based on sensor data collected during production.

Virtual metrology (VM) uses machine-learning models to predict post-process wafer properties, such as layer thickness, from routinely collected process and sensor data. These predictions can reduce the need for slow, costly, or destructive physical measurements.

As fewer wafers are physically measured, **thickness labels become sparse**. Consequently, the standard accuracy metrics used to monitor the VM model cannot be computed routinely.

How can we continuously monitor deployed VM predictions between physical thickness measurements?

2. Drift detection cannot confirm accuracy

Existing drift-detection methods monitor whether sensor inputs or model predictions have changed relative to the training period. However, a distributional change does not necessarily indicate inaccurate predictions, and the absence of drift does not confirm that the model remains accurate.

Between physical thickness measurements, drift detection cannot provide an **independent physical check of the VM predictions**.

3. Resistance provides a physical check

Although VM substitutes for costly thickness measurements, **sheet resistance remains routinely measured** in production.

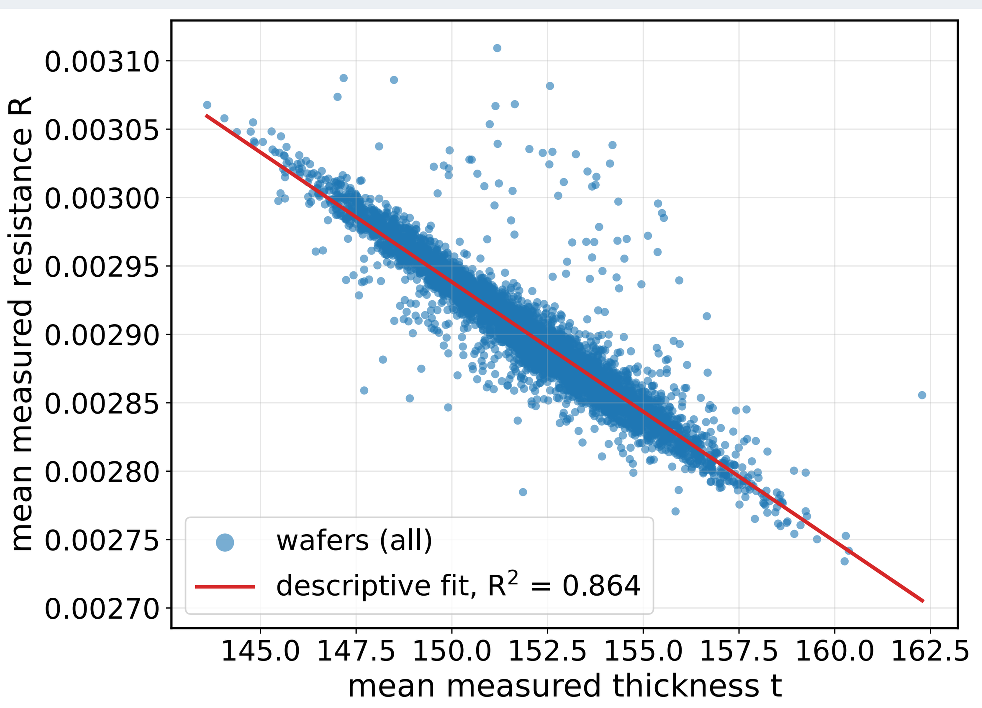
For a deposited layer with approximately stable resistivity, sheet resistance is inversely related to thickness:

$$R_s \approx \frac{\rho}{t}.$$

Over the observed operating range, this relationship can be calibrated from historical measurements. A VM-predicted thickness then implies an expected sheet resistance:

$$R_{\mathrm{exp}} = g(t_{\mathrm{pred}}) \approx a\, t_{\mathrm{pred}} + b.$$

Comparing expected and measured resistance provides a physically grounded check between thickness measurements.



F2. Measured wafer-mean sheet resistance decreases as thickness increases. The fitted relationship converts VM-predicted thickness into the expected resistance used for physics-consistency monitoring.

Contact and support

HCAI Sarajevo
Amina Mević
amevic@etf.unsa.ba

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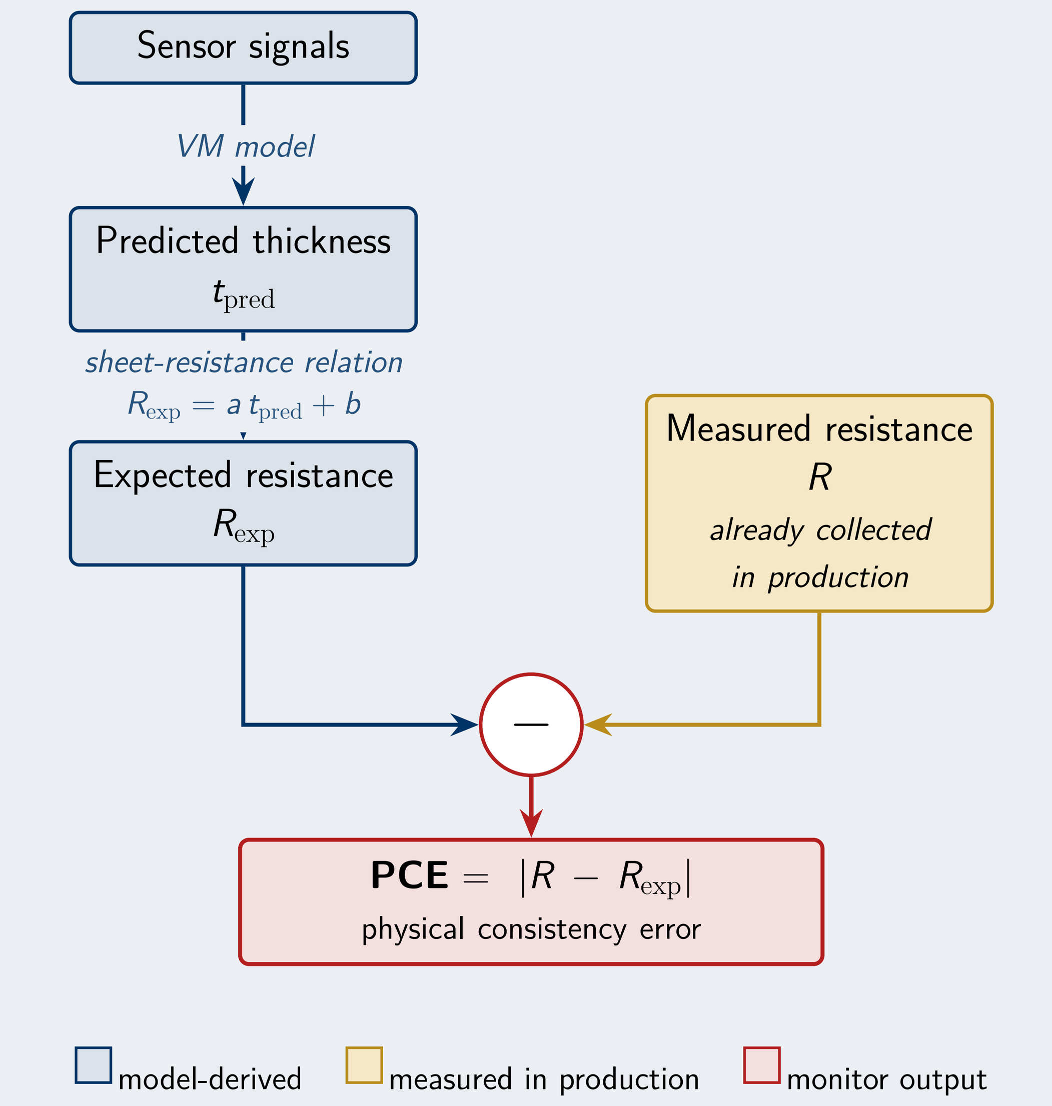
Proposed method: two views of each wafer



Physics-consistency metric family

Metric	Definition	Interpretation
PCE	$\mathrm{PCE}_i = \lvert R_i - R_{\mathrm{exp},i}\rvert$	Absolute discrepancy in resistance units
NPCE	$\mathrm{NPCE}_i = \dfrac{\mathrm{PCE}_i}{\sigma_R}$	Discrepancy relative to the spread of measured resistance
RPE	$\mathrm{RPE}_i = \dfrac{\mathrm{PCE}_i}{\lvert R_i\rvert}$	Discrepancy relative to the measured resistance level

Here, σ_R is the standard deviation of measured resistance in the evaluated monitoring window. Metrics are summarised across wafers using their mean or distribution.

Experimental setup

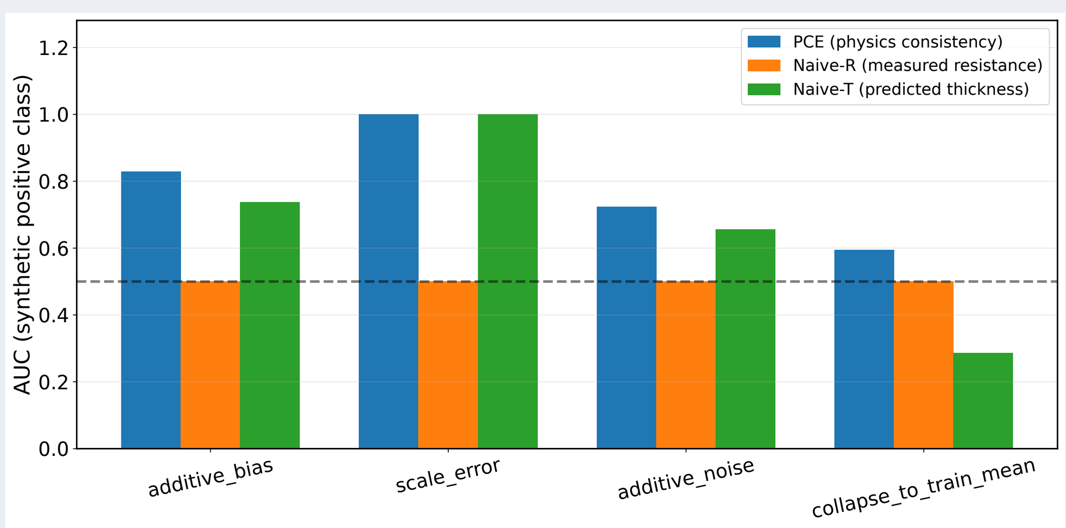
- **Data:** 5,141 wafers with process signals, thickness, and sheet-resistance measurements.
- **Controlled degradation:**
  - additive prediction drift,
  - calibration (scale) error,
  - additive prediction noise,
  - collapse towards the training mean.

4. Two complementary views of model performance

Model	Thickness accuracy $R^2(t, t_{\mathrm{pred}})$	Resistance agreement $R^2(R, R_{\mathrm{exp}})$
Decision tree	0.435 ± 0.020	0.344 ± 0.029
Linear (Ridge)	0.605 ± 0.014	0.538 ± 0.023
Random forest	0.685 ± 0.013	0.580 ± 0.022
XGBoost	$\mathbf{0.717 \pm 0.009}$	$\mathbf{0.620 \pm 0.018}$

T1. Mean ± standard deviation over five random hold-outs.

5. PCE reveals degradation missed by single-signal monitors

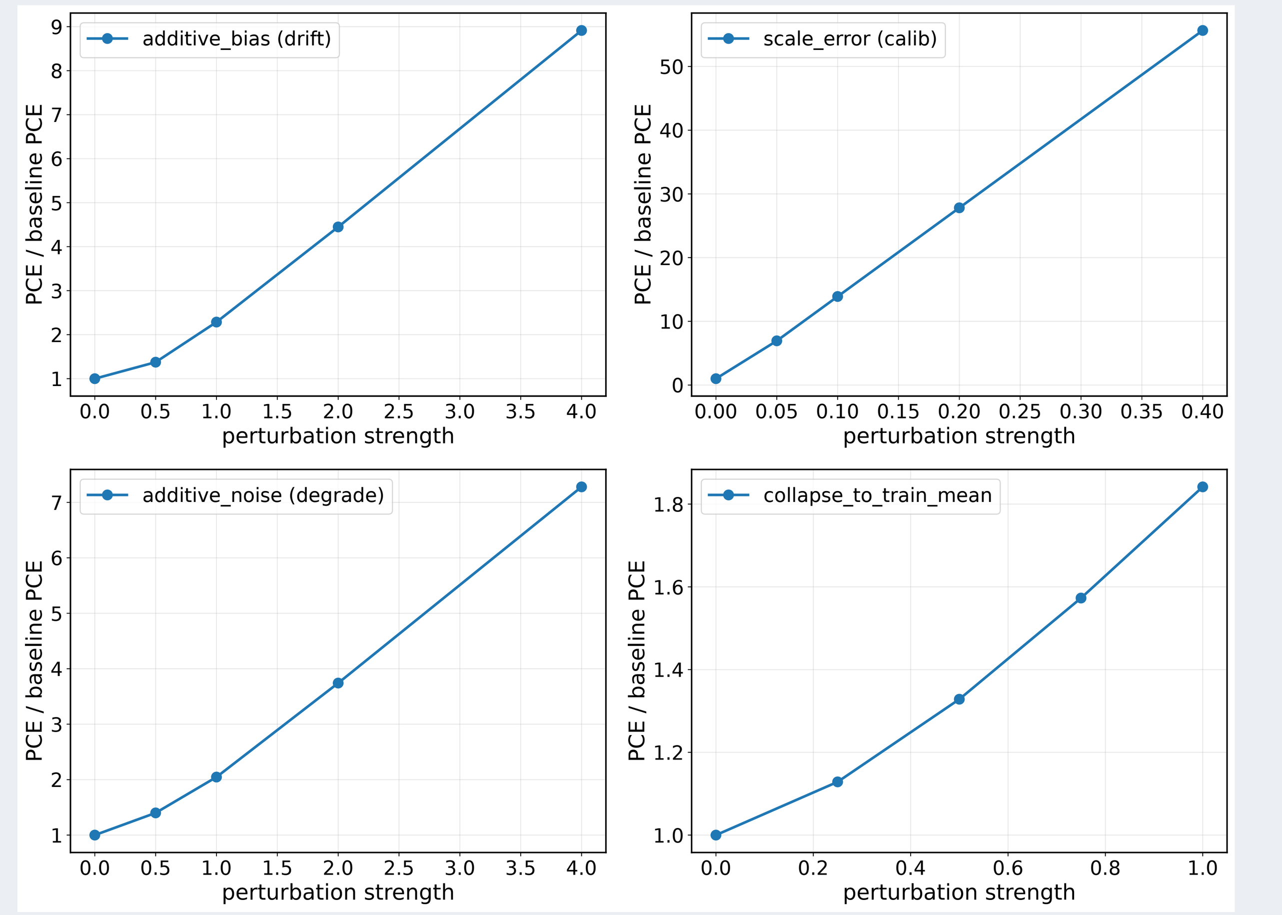


F3. Ability to distinguish unperturbed from synthetically degraded VM predictions.

Three label-free signals are tested on the same controlled perturbations:

- **Naive-R (measured resistance)** cannot detect changes applied only to VM predictions ($\mathrm{AUC} = 0.5$).
- **Naive-T (predicted thickness)** detects additive bias and scale error, but responds in the wrong direction when predictions collapse towards the training mean ($\mathrm{AUC} = 0.286$).
- **PCE (physics consistency)** combines predicted thickness with measured resistance and provides above-chance separation for all four degradation modes.

6. PCE increases with degradation strength



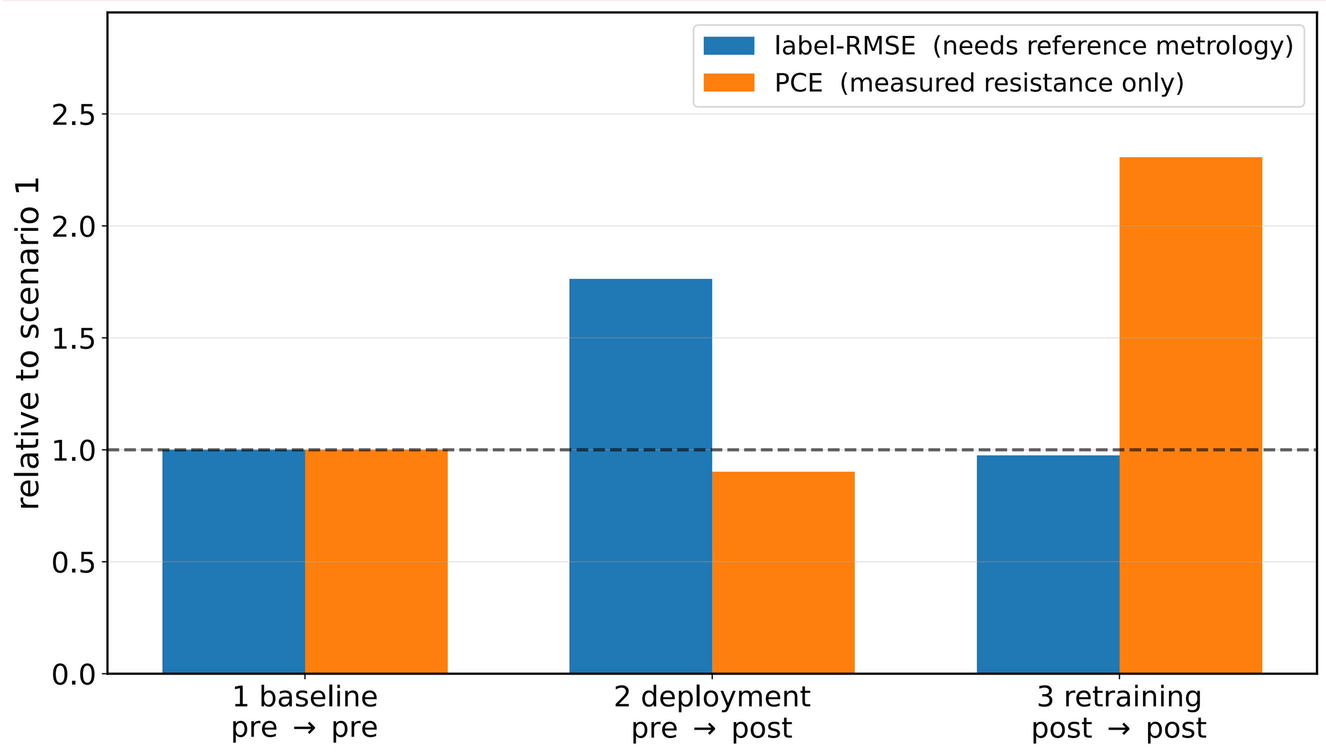
F4. PCE relative to its unperturbed baseline across increasing strengths of four controlled prediction perturbations. A value of 1 denotes baseline performance.

Degradation mode	Maximum PCE increase	Detection rate
Additive bias	$\times 8.38 \pm 0.33$	32 %
Scale error	$\times\mathbf{52.6 \pm 2.0}$	**100 %**
Additive noise	$\times 6.82 \pm 0.18$	29 %
Collapse to train mean	$\times 1.79 \pm 0.05$	10 %

T2. Maximum PCE increase: mean ± standard deviation over five random hold-outs. Detection rate: one representative perturbation strength at a 5 % false-alarm rate, evaluated on a single hold-out.

7. Real-world case: PCE challenges label-driven retraining

F5. Retrospective view of 5,749 production wafers from 2021–2025. The effective-resistivity proxy $R \cdot t$ shows a distinct change around June 2025.



F6. Label-RMSE and PCE across baseline, deployment after the observed shift, and retraining on post-shift thickness labels. Both metrics are shown relative to the pre-event baseline.

Industrial benefit: RMSE indicates that the retrained model fits the post-event thickness labels, whereas PCE reveals that its predictions are no longer consistent with the independently measured resistance. This discrepancy triggers metrology and process verification before the retrained model is approved for deployment.

Take-home message

- Routine resistance enables **label-free physical-consistency monitoring** between thickness measurements.
- PCE increased under all tested perturbations and by $\times 2.31$ **after retraining** in the production case.
- PCE flags inconsistency, **but does not directly measure accuracy or identify its cause**.

Good accuracy does not guarantee physical consistency.

References

[1] Maitra et al., "Virtual metrology in semiconductor manufacturing," *Expert Systems with Applications*, 2024.
[2] Li and Hsieh, "Concept drift detection and adaptation in virtual metrology," *IEEE Access*, 2025.
[3] Kao et al., "Run-to-run control utilizing virtual metrology with reliance index," *IEEE Trans. Semiconductor Manufacturing*, 2013.
[4] Mayadas and Shatzkes, "Electrical resistivity of polycrystalline films," *Physical Review B*, 1970.